Racial Disparities in Traffic Stops: Orange County Sheriff's Department (2020)

RIPA Stop Data Analysis

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Abstract

This paper studies racial disparities in traffic stops conducted by the Orange County Sheriff's Department in 2020 using data collected under the California Racial and Identity Profiling Act (RIPA). I first document racial differences in stop volumes, stop reasons, and search rates. I then apply the Veil of Darkness test to examine whether minority drivers are more likely to be stopped when officers can observe race prior to the stop. Finally, I conduct an outcome (hit-rate) test to assess whether searches of minority drivers are less productive. The combined evidence suggests meaningful racial disparities in policing outcomes, though each test has limitations that preclude a fully causal interpretation.

1 Context and Summary Statistics

Data

I use 2020 RIPA stop data from the Orange County Sheriff's Department.[1] Orange County is a large, suburban county with a mix of incorporated cities and unincorporated areas policed by the Sheriff's Department.

The raw dataset includes all reported traffic stops conducted during 2020. I restrict the sample to traffic stops with real values for perceived race, and valid stop timestamps. These restrictions ensure comparability across racial groups and consistency across analyses.

Task A

Table 1 reports total stops by race, the racial composition of all stops, and a few interesting reasons for stopping: moving violations, equipment violations, and investigative stops. These categories differ in the degree of officer discretion involved.

[1] `https://openjustice.doj.ca.gov/data`

Table 1: Stop Characteristics by Race, Orange County Sheriff's Department (2020)

	Total Stops	% of Stops	Moving (%)	Equipment (%)	Investigative (%)
Asian	2520	6.8	0.0	0.0	2.4
Black	1585	4.3	0.0	0.0	8.8
Hispanic	13 612	36.7	0.0	0.0	4.8
White	19 423	52.3	0.0	0.0	5.2
Total	37 140	100.0	0.0	0.0	5.1
Total		100.0			

Task B

I define a search indicator to be if any search occurred during the stop, including consent, probable cause, incident-to-arrest, or warrant searches. This measure captures overall search behavior without differentiating by search type. Table 2 shows the search rates by race.

Table 2: Person Search Rates by Race/Ethnicity (Orange County, 2020)

	Search Rate
Hispanic / Latino	0.251
White	0.199
Black / African American	0.225
Asian	0.104
Other / Multiracial	0.240

Does a Disparity Prove Discrimination?

Differences in search rates across racial groups are consistent with racial discrimination but cannot alone establish discriminatory intent. Officers may be responding to underlying factors that are correlated with race that would not be recorded in the data.

Another complication is the benchmark problem: racial distribution of drivers on the road is not observed. Without knowing the appropriate comparison population, raw disparities cannot be interpreted causally. The distribution of race among drivers can be driven by factors such as income that could dictate car ownership/use. These limitations motivate an attempt to keep officer information constant. The next sections apply the Veil of Darkness test and an outcome test to address this.

2 The Veil of Darkness Test

Design

The Veil of Darkness test exploits seasonal variation in sunset times. In theory, during daylight, officers can more effectively perceive the driver's race prior to the stop, whereas after dark this information is unavailable. If minority drivers are stopped at higher rates in daylight than in darkness, we can more accurately conclude that discrimination is occurring.

I restrict attention to a time window around sunset and include 15-minute fixed effects to flexibly control for variation in traffic patterns over short intervals. This helps ensure that comparisons between daylight and darkness are made among drivers observed at similar times of day. Sunset timing variation across days is used to estimate exogenous variation across visibility conditions.

Sample Restriction

The VOD sample includes only stops occurring between 5:00 and 8:30 PM. This window contains overlap between daylight and darkness across all seasons.

Specification

The Veil of Darkness Regression estimates the effect of daylight on the probability that a stopped driver is perceived as Black or Hispanic.

Table 3: Veil-of-Darkness Regression: Effect of Daylight on Minority Stop Rate

	(1) Black or Hispanic	(2) Black	(3) Hispanic
Daylight	0.0948	0.0047	0.0892
	(0.0239)	(0.0099)	(0.0236)
15-min dummies	Yes	Yes	Yes
Observations	5212	5212	5212

Positive coefficients indicate a higher likelihood of minority stops during daylight conditions relative to darkness.

Does the probability of stopping a minority driver rise when the sun is up? What does this mean, and what does it not mean?

A positive daylight coefficient indicates that minority drivers constitute a relatively greater share of stops when officers can observe race (light out). This result is consistent with racial profiling in all 3 cases but does not identify specific officer behavior. The validity of this test hinges on the assumption that the racial composition of drivers does not change between daylight and darkness within the specified window (5-8:30 PM). If driver travel patterns differ by lighting conditions in

ways correlated with race, the assumption would fail, and the estimate would be biased. However, within a narrow time window around sunset, we can assume such differences to be minimal, making the Veil of Darkness test a meaningful approach in detecting potential discrimination.

3 The Outcome Test: Hit Rates

Design

The outcome test compares the success rate of searches across racial groups. If officers apply lower suspicion thresholds to minority drivers, searches of those drivers should be less likely to uncover contraband. I define "any contraband" as the discovery of drugs, weapons, or other illegal items.

Specification

The regression is estimated on the subsample of stops involving a search, with white drivers as the omitted category.

Table 4: Hit-Rate Regression and Search/Contraband Rates by Race

	(1) Contraband	Search Rate (%)	Hit Rate (%)
White	0.0000	20.8	37.1
Black	-0.0690	24.5	29.6
Hispanic	-0.0468	27.2	31.8

How different are search rates by race compared to hit rates? What does this tell you about the potential for disparate policing? What are the limitations of this test?

If groups with higher search rates also exhibit lower hit rates, this pattern is consistent with the possibility that lower thresholds for suspicion are applied to those groups, implying disparate policing. However, this interpretation is not concrete and the test is limited. Differences in hit rates could also occur because of heterogeneous distributions of contraband risk across racial groups, even with equal search thresholds. As a result, the outcome test cannot isolate discrimination at the margin of search decision.

4 Python Code

```
import pandas as pd
import numpy as np

pd.set_option("display.float_format", "{:.3f}".format)
```

```
df = pd.read_csv("RIPAoc2020.csv", low_memory=False)

# basic cleaning
df["DATE_OF_STOP"] = pd.to_datetime(df["DATE_OF_STOP"], errors="coerce")
df = df[df["DATE_OF_STOP"].dt.year == 2020].copy()

# ensure stop reason is string
df["REASON_FOR_STOP"] = df["REASON_FOR_STOP"].astype(str).str.upper()


race_cols = {
    "White": "RAE_WHITE",
    "Black": "RAE_BLACK_AFRICAN_AMERICAN",
    "Hispanic": "RAE_HISPANIC_LATINO",
    "Asian": "RAE_ASIAN",
    "Other/Unknown": "RAE_MULTIRACIAL"
}

for c in race_cols.values():
    df[c] = pd.to_numeric(df[c], errors="coerce")

def get_race(row):
    for race, col in race_cols.items():
        if row[col] == 1:
            return race
    return np.nan

df["RACE"] = df.apply(get_race, axis=1)
df = df[df["RACE"].notna()].copy()


rs_cols = [
    "RFS_RS_REASON_SUSP",
    "RFS_RS_VIOLENT_CRIME",
    "RFS_RS_DRUG_TRANS",
    "RFS_RS_MATCH_SUSPECT"
]
```

```
for c in rs_cols:
    df[c] = pd.to_numeric(df[c], errors="coerce").fillna(0)

df["MOVING"] = df["REASON_FOR_STOP"].str.contains("MOVING", na=False)
df["EQUIPMENT"] = df["REASON_FOR_STOP"].str.contains("EQUIP", na=False)

df["INVESTIGATIVE"] = (
    (df["RFS_RS_REASON_SUSP"] == 1) |
    (df["RFS_RS_VIOLENT_CRIME"] == 1) |
    (df["RFS_RS_DRUG_TRANS"] == 1) |
    (df["RFS_RS_MATCH_SUSPECT"] == 1)
)


table1 = df.groupby("RACE").agg(
    total_stops=("RACE", "size"),
    moving=("MOVING", "mean"),
    equipment=("EQUIPMENT", "mean"),
    investigative=("INVESTIGATIVE", "mean")
)

table1[["moving", "equipment", "investigative"]] *= 100
table1["pct_stops"] = table1["total_stops"] / len(df) * 100

table1 = table1[[
    "total_stops",
    "pct_stops",
    "moving",
    "equipment",
    "investigative"
]]

table1.columns = [
    "Total Stops",
    "% of Stops",
    "Moving (%)",
    "Equipment (%)",
    "Investigative (%)"
]
```

```
table1


total_row = pd.DataFrame({
    "Total Stops": [len(df)],
    "% of Stops": [100.0],
    "Moving (%)": [df["MOVING"].mean() * 100],
    "Equipment (%)": [df["EQUIPMENT"].mean() * 100],
    "Investigative (%)": [df["INVESTIGATIVE"].mean() * 100],
}, index=["Total"])

table1_final = pd.concat([table1, total_row])

table1_final


df_vod = df.copy()
df_vod["TIME_OF_STOP"] = pd.to_numeric(df_vod["TIME_OF_STOP"], errors="coerce")

df_vod = df_vod[df_vod["TIME_OF_STOP"].notna()].copy()

df_vod["HOUR"] = (df_vod["TIME_OF_STOP"] // 100).astype(int)
df_vod["MINUTE"] = (df_vod["TIME_OF_STOP"] % 100).astype(int)

df_vod = df_vod[(df_vod["MINUTE"] < 60) & (df_vod["HOUR"] <= 23)].copy()


df_vod = df_vod[
    (df_vod["HOUR"] >= 17) &
    (df_vod["HOUR"] <= 20)
].copy()


df_vod["BLACK"] = pd.to_numeric(df_vod["RAE_BLACK_AFRICAN_AMERICAN"], errors="coerce").fillna(0
df_vod["HISP"] = pd.to_numeric(df_vod["RAE_HISPANIC_LATINO"], errors="coerce").fillna(0)

df_vod["MINORITY"] = ((df_vod["BLACK"] == 1) | (df_vod["HISP"] == 1)).astype(int)
```

```
df_vod["TIME_BIN"] = df_vod["HOUR"] * 4 + (df_vod["MINUTE"] // 15)

df_vod["DAYLIGHT"] = (df_vod["HOUR"] < 19).astype(int)


import statsmodels.formula.api as smf

m1 = smf.ols("MINORITY ~ DAYLIGHT + C(TIME_BIN)", data=df_vod).fit(cov_type="HC1")
m2 = smf.ols("BLACK ~ DAYLIGHT + C(TIME_BIN)", data=df_vod).fit(cov_type="HC1")
m3 = smf.ols("HISP ~ DAYLIGHT + C(TIME_BIN)", data=df_vod).fit(cov_type="HC1")

print(len(df_vod))


results_table = pd.DataFrame({
    "Model": ["Black or Hispanic", "Black", "Hispanic"],
    "Daylight Coef": [
        m1.params["DAYLIGHT"],
        m2.params["DAYLIGHT"],
        m3.params["DAYLIGHT"]
    ],
    "Std. Error": [
        m1.bse["DAYLIGHT"],
        m2.bse["DAYLIGHT"],
        m3.bse["DAYLIGHT"]
    ],
    "t-Statistic": [
        m1.tvalues["DAYLIGHT"],
        m2.tvalues["DAYLIGHT"],
        m3.tvalues["DAYLIGHT"]
    ],
    "P-Value": [
        m1.pvalues["DAYLIGHT"],
        m2.pvalues["DAYLIGHT"],
        m3.pvalues["DAYLIGHT"]
    ],
    "Observations": [
        int(m1.nobs),
```

```
        int(m2.nobs),
        int(m3.nobs)
    ]
})

results_table


df_out = df.copy()
df_out["ADS_SEARCH_PERSON"] = pd.to_numeric(df_out["ADS_SEARCH_PERSON"], errors="coerce").fill
df_out["ADS_SEARCH_PROPERTY"] = pd.to_numeric(df_out["ADS_SEARCH_PROPERTY"], errors="coerce").

df_out["ANY_SEARCH"] = ((df_out["ADS_SEARCH_PERSON"] == 1) | (df_out["ADS_SEARCH_PROPERTY"] ==


contraband_cols = [
    "CED_FIREARM",
    "CED_AMMUNITION",
    "CED_WEAPON",
    "CED_DRUGS",
    "CED_ALCOHOL",
    "CED_MONEY",
    "CED_DRUG_PARAPHERNALIA",
    "CED_STOLEN_PROP",
    "CED_ELECT_DEVICE",
    "CED_OTHER_CONTRABAND"
]

for c in contraband_cols:
    df_out[c] = pd.to_numeric(df_out[c], errors="coerce").fillna(0)

df_out["CONTRABAND"] = (df_out[contraband_cols].sum(axis=1) > 0).astype(int)


race_cols = {
    "White": "RAE_WHITE",
    "Black": "RAE_BLACK_AFRICAN_AMERICAN",
    "Hispanic": "RAE_HISPANIC_LATINO",
    "Asian": "RAE_ASIAN",
```

```
    "Other": "RAE_MULTIRACIAL"
}

for c in race_cols.values():
    df_out[c] = pd.to_numeric(df_out[c], errors="coerce").fillna(0)

def assign_race(row):
    if row["RAE_WHITE"] == 1:
        return "White"
    elif row["RAE_BLACK_AFRICAN_AMERICAN"] == 1:
        return "Black"
    elif row["RAE_HISPANIC_LATINO"] == 1:
        return "Hispanic"
    elif row["RAE_ASIAN"] == 1:
        return "Asian"
    elif row["RAE_MULTIRACIAL"] == 1:
        return "Other"
    else:
        return np.nan

df_out["RACE"] = df_out.apply(assign_race, axis=1)

df_out = df_out[df_out["RACE"].notna()].copy()


df_hit = df_out[df_out["ANY_SEARCH"] == 1].copy()
print(f"Number of hits: {len(df_hit)}")


model_hit = smf.ols("CONTRABAND ~ C(RACE)", data=df_hit).fit(cov_type="HC1")
model_hit.summary()


summary = df_out.groupby("RACE").apply(
    lambda g: pd.Series({
        "Search Rate (%)": g["ANY_SEARCH"].mean() * 100,
        "Hit Rate (%)": g.loc[g["ANY_SEARCH"] == 1, "CONTRABAND"].mean() * 100 if g["ANY_SEARC
    })
).reset_index()
```

```
summary


coef_table = pd.DataFrame({
    "Race": model_hit.params.index,
    "Coefficient": model_hit.params.values,
    "Std. Error": model_hit.bse.values,
    "p-value": model_hit.pvalues.values
})

coef_table


print("Total observations:", len(df_out))
print("Sample size:", len(df_hit))
```